



A comprehensive survey on 6G: Enabling technologies, key applications, and future challenges

Sofiane Latreche^{ID}*, Hocine Bellahsene^{ID}

Université de Bejaia, Faculté de Technologie, Laboratoire de Génie Electrique, 06000 Bejaia, Algeria

ARTICLE INFO

Keywords:

5G/6G
Emerging technologies
Wireless communication
Mobile communication
Survey

ABSTRACT

The rapid evolution of wireless communication technologies from 1G to 5G has enabled remarkable advances in connectivity, yet emerging applications such as holographic telepresence, autonomous systems, wireless brain-computer interfaces, and ubiquitous 3D coverage expose fundamental limitations in current networks. While numerous surveys have explored individual 6G enablers, few studies provide a unified and comprehensive analysis that integrates the technological foundations, architectural transformations, and multidisciplinary applications required to realize future 6G systems. This manuscript addresses this gap by offering a holistic overview of 6G, beginning with the motivation for moving beyond 5G and the performance demands driving next-generation research. It synthesizes key enabling technologies including terahertz communication, ultra-massive MIMO, intelligent reflective surfaces, AI-native networking, Digital Twin Networks, and generative artificial intelligence. The paper further examines new architectural models spanning terrestrial, aerial, satellite, and underwater networks, while also discussing emerging 6G use cases across healthcare, robotics, industry, mobility, and immersive environments. Finally, the manuscript outlines open challenges and research directions that must be addressed to transform the 6G vision into practical reality. By bridging foundational concepts and forward-looking innovations, this survey provides a comprehensive reference for researchers and practitioners seeking to understand and contribute to the evolving 6G landscape.

1. Introduction

The evolution of mobile communication networks has been one of the most transformative technological journeys of the modern era [1]. Since the deployment of the first generation (1G) cellular networks in the 1980s, each subsequent generation, 2G, 3G, 4G, and more recently 5G, has introduced significant advancements in data rate, latency, network capacity, and service capabilities. These generational shifts have not only enhanced user experience but also catalyzed the proliferation of mobile-based services and digital ecosystems. Driven by rapid advancements in hardware, spectrum utilization, and protocol design, the mobile network landscape has continuously evolved to meet the ever-growing demand for connectivity and mobility in both personal and industrial domains [2].

Fifth-generation (5G) networks represent a considerable leap forward in wireless communication, introducing novel features such as enhanced Mobile Broadband (eMBB), Ultra-Reliable and Low-Latency Communications (URLLC), and Massive Machine-Type Communications (mMTC) [3]. These features have supported a wide range of applications including augmented and virtual reality, smart city infrastructures, remote healthcare, and Industry 4.0 paradigms. However,

despite these capabilities, 5G networks are increasingly facing limitations in addressing the demands of next-generation use cases that require ultra-high reliability, extremely low latency, unprecedented data rates, and intelligent, context-aware networking capabilities [4].

The projected demands of the 2030s, such as holographic telepresence, tactile internet, wireless brain-computer interfaces, and the integration of artificial intelligence (AI) into every layer of the network, have underscored the necessity for a new paradigm in mobile communications [5]. The sixth generation (6G) is envisioned to transcend the foundational capabilities of 5G by incorporating Terahertz (THz) communications, intelligent reflective surfaces, quantum and blockchain technologies, AI-native network management, and non-terrestrial networks (NTNs), thereby enabling global, three-dimensional connectivity with seamless interoperability [6]. The development of 6G aims not only to address the shortcomings of current systems but also to anticipate future societal, industrial, and technological demands in a sustainable and inclusive manner [7].

The contributions of this work can be summarized across several key dimensions. First, the manuscript provides a comprehensive motivation for the shift beyond 5G, clearly outlining the current limitations of

* Corresponding author.

E-mail addresses: sofiane.latreche@univ-bejaia.dz (S. Latreche), hocine.bellahsene@univ-bejaia.dz (H. Bellahsene).

Table 1

Comparison of representative 6G surveys and their relation to this work.

Survey	Year	Scope/Focus	Limitations compared to this work
Huang et al. [8]	2019	Green 6G networks, energy-efficient design, early architecture concepts.	Focuses mainly on sustainability; does not cover modern enablers such as DTN, generative AI, or concrete use-case workflows.
Uusitalo et al. (Hexa-X) [9]	2021	European 6G vision, key use cases, high-level enabling technologies.	Visionary but conceptual; lacks detailed enabling-technology comparisons and practical 6G application demonstrations.
Bassoli et al. [10]	2021	Motivation for 6G and limitations of 5G; early research drivers.	Strong on motivation but limited coverage of technologies, architectures, and integrated 6G system design.
Salameh & El Tarhuni [11]	2022	Overview of 5G→6G evolution, challenges, and emerging technologies.	Limited architectural depth and does not integrate AI-native networks, DTN, or practical mission-critical workflows.
Hakeem et al. [12]	2022	Vision and research directions; identifies challenges and enablers.	Does not provide a unified comparison of technologies, architectures, and detailed real-world scenarios.
Shafi et al. [2]	2024	High-level evolution towards 6G; key performance indicators and trends.	Technology review remains broad; does not explore integrated architectures or detailed use-case implementations.
This work	2025	Comprehensive survey covering 6G technologies, architectures, applications, DTN, GenAI, and practical workflows.	Provides unified analysis of enabling technologies, detailed 3D architectures, realistic holographic-surgery scenario, and clear research directions, offering greater breadth and depth than prior surveys.

existing networks and the emerging demands that justify the need for 6G. Building on this foundation, the survey offers a holistic and structured review of essential 6G enabling technologies, including terahertz communication, ultra-massive MIMO, intelligent reflective surfaces, AI-native networking, edge intelligence, Digital Twin Networks, and generative artificial intelligence, emphasizing their roles, advantages, and open research challenges. In addition, the paper delivers a detailed analysis of next-generation network architectures by presenting an integrated perspective of space–air–ground–sea systems, highlighting architectural design principles, major components, interoperability constraints, and system-level considerations. The manuscript further examines a broad set of emerging 6G application domains such as extended reality, autonomous systems, intelligent healthcare, the Internet of Everything, and holographic communication. To illustrate the practical relevance of these technologies, a realistic end-to-end scenario of 6G-enabled remote holographic surgery is presented, demonstrating how multiple 6G enablers converge to support mission-critical operations. Finally, the survey synthesizes the major technical challenges and research opportunities associated with the development of 6G networks, identifying open problems and outlining promising directions for future investigation. Taken together, these contributions position this work as a comprehensive and forward-looking reference for both academic researchers and industry practitioners engaged in shaping the next generation of wireless communication systems.

This paper presents a comprehensive survey of 6G mobile networks, focusing on the technological trajectory that has led to its emergence, and the fundamental requirements it seeks to fulfill. The paper is structured as follows: Section 2 provides an overview of the historical evolution of mobile networks from 1G to 5G, highlighting the key advancements and limitations that have motivated the shift towards 6G. Section 3 outlines the technological motivations and conceptual vision of 6G, including performance objectives and expected capabilities. Section 4 introduces and analyzes the enabling technologies that will form the backbone of 6G infrastructure, such as Terahertz communication, edge intelligence, massive MIMO, and quantum-enhanced security frameworks. Section 5 discusses the anticipated changes in core network architecture, emphasizing the need for a flexible, scalable, and intelligent design. Section 6 explores the diverse use cases and application scenarios that 6G aims to support, ranging from immersive extended reality environments to autonomous systems and intelligent healthcare. Section 7 outlines the major technical and theoretical challenges that must be addressed to realize the full potential of 6G networks. Finally, the concluding section summarizes the key insights and offers directions for future research. This structure is illustrated in Fig. 1.

By synthesizing current research trends and future projections, this paper aims to provide a holistic understanding of the forthcoming 6G era and to contribute to the ongoing academic and industrial discourse on next-generation mobile communications.

Although several survey papers have examined different aspects of 6G, their coverage varies significantly in terms of scope, depth, and technological focus. Table 1 summarizes these works and clearly demonstrates how this manuscript provides a broader and more integrated perspective, covering enabling technologies, architectures, applications, and practical use-case workflows.

2. Mobile network evolution

Mobile networks have undergone a remarkable and rapid evolution since their inception in the 1980s. This continuous evolution has given rise to several generations of mobile telephony, with each succeeding generation being introduced approximately every decade [13].

Fig. 2 provides a visual representation of the progression of mobile network evolution.

The Table 2 presents a comparative overview of mobile network generations, highlighting their main features, technologies and applications.

2.1. First steps from 1G to 3G

The inception of cellular networks dates back to the 1980s, when the first generation was introduced, primarily designed for voice services and capable of data rates up to 2.4 kbps. In contrast to the analog systems of the first generation, the second generation embraced digital modulation technologies like Time Division Multiple Access (TDMA). This technological shift brought about a significant enhancement, offering data rates of up to 64 kbps and introducing features such as short message service (SMS) [14]. The dawn of the third generation in the year 2000 marked a central moment in mobile network evolution, aiming to provide high-speed data transmission. The 3G network was a game-changer, offering a minimum data transfer rate of 2 Mbps and granting users access to the vast realm of the internet. This transformation enabled an array of advanced services previously unsupported by the 1G and 2G networks, including web browsing, TV streaming, navigation maps, and video services [15].

2.2. 4G evolution

The advent of 4G marked a significant milestone in the evolution of mobile networks, introduced in the late 2010s [13]. This all-IP network

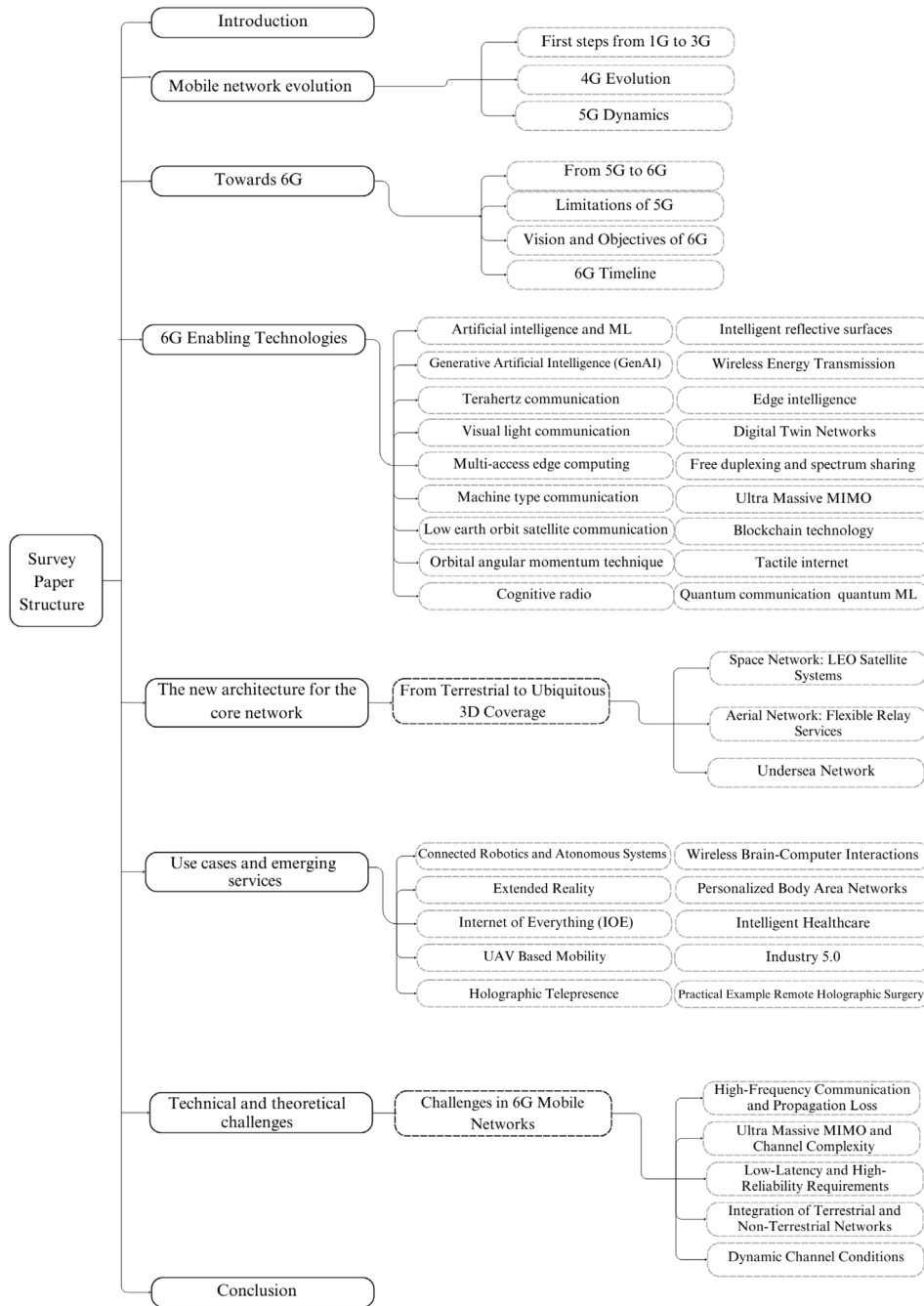


Fig. 1. Structure of the paper with main sections and flow.

was designed to deliver exceptionally high-speed data rates, reaching up to 1 Gbps for downlink and 500 Mbps for uplink transmissions. Its core enhancements included improvements in spectral efficiency and a reduction in latency, making it well-suited to meet the requirements of demanding applications such as digital video broadcasting (DVB), high-definition TV content delivery, and seamless video chat experiences [6]. Notably, Long Term Evolution-Advanced (LTE-A) and Wireless Interoperability for Microwave Access (WiMax) were both recognized as 4G standards. The foundation of 4G, LTE, integrated a blend of existing and novel technologies, including Coordinated Multi-Point (CoMP), multiple input multiple output (MIMO), and orthogonal frequency division multiplexing (OFDM) [16].

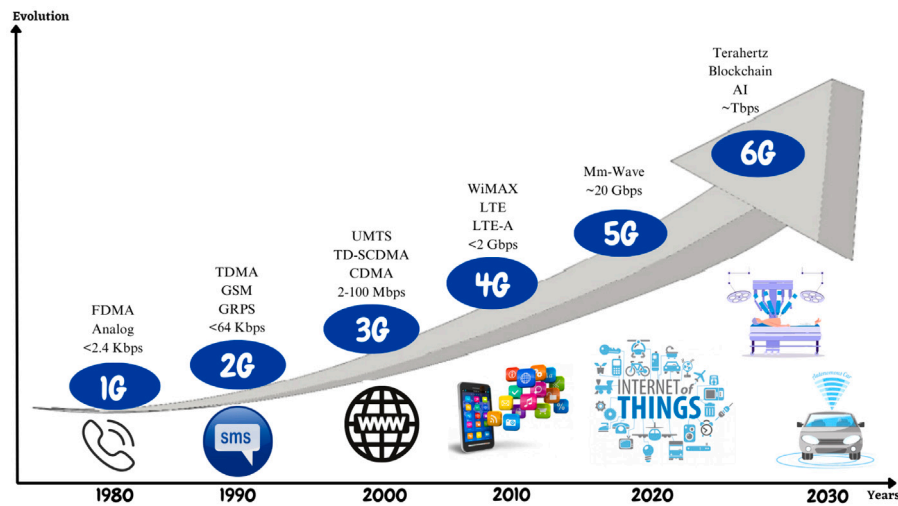
2.3. 5G dynamics

The primary objective of 5G technology is to achieve revolutionary advances in terms of throughput, latency, network reliability, and energy efficiency, all while ensuring extensive connectivity. Unlike its predecessors, 5G harnesses not only the new sub-6 GHz spectrum, which includes the C-band and the broader 6 GHz band but also, for the first time, delves into the millimeter-wave band, enabling significantly higher data rates, reaching up to an impressive ten Gbps [18]. 5G relies on advanced access technologies, including beam division multiple access (BDMA), real-time division multiple access, and filter bank multiple access (FBMC). To enhance network performance, 5G incorporates

Table 2

Comparison of mobile network generations from 1G to 6G.

Feature	1G	2G	3G	4G	5G	6G
Introduction	1980s	1990s	2000s	2010s	2020s	2030s (Expected)
Technology	Analog	Digital	Packet-Switched	LTE (IP-based)	NR (New Radio)	AI-driven, THz frequencies
Data rate	2.4 kbps	64 kbps–200 kbps	2 Mbps–42 Mbps	100 Mbps–1 Gbps	10 Gbps	1 Tbps (Expected)
Latency	High (600 ms)	300–600 ms	100–500 ms	10–50 ms	1–10 ms	<1 ms
Multiple access	FDMA	TDMA, CDMA	CDMA, WCDMA	OFDMA, SC-FDMA	OFDMA, mmWave	AI-Driven, Reconfigurable Intelligent Surfaces (RIS)
Key features	Voice calls	SMS, Basic data	Mobile internet, Video calls	High-speed internet, VoLTE	Ultra-reliable, Low latency, Massive IoT	AI-Integrated, Quantum communication
Frequency bands	800 MHz	900 MHz, 1800 MHz	850 MHz–2100 MHz	700 MHz–2.5 GHz	Sub-6 GHz, mmWave (24–100 GHz)	THz band (>100 GHz)
Main applications	Voice communication	SMS, Basic internet	Mobile web, Video streaming	HD streaming, IoT, VoLTE	Smart cities, IoT, Industry 4.0	Holographic communication, Tactile internet, Smart implants
Challenges	Poor security, Low capacity	Limited data rate	Spectrum congestion	Battery drain, Network congestion	Infrastructure cost, Deployment complexity	Energy efficiency, THz signal penetration

**Fig. 2.** Evolution of mobile wireless systems [11].

a multitude of emerging technologies. These innovations encompass massive MIMO to boost network capacity, software-defined networks (SDN) for increased network flexibility, device-to-device (D2D) communication to improve spectral efficiency, information-centric networking (ICN) to reduce network traffic, and network slicing to facilitate the swift deployment of numerous services [17].

Within the framework of IMT 2020, three primary usage scenarios have been proposed for 5G technology: enhanced mobile broadband (eMBB) to cater to high-speed data requirements, ultra-reliable low-latency communications (URLLC) for mission-critical applications, and massive Machine Type Communications (mMTC) to support the proliferation of connected devices and the Internet of Things like presented in Fig. 3.

3. Towards 6G

3.1. From 5G to 6G: Why do we need a new generation?

The evolution from 5G to 6G is driven by the ever-increasing demands for higher speeds, lower latencies, and greater connectivity

density. While 5G has brought unprecedented capabilities, such as enhanced mobile broadband (eMBB), ultra-reliable low-latency communications (URLLC), and massive machine-type communications (mMTC), the growing complexity of applications in industries, health, education, and entertainment has revealed the need for an even more advanced network [19]. Applications such as holographic telepresence, and extended reality (XR) are expected to require terabit-per-second (Tbps) data rates and latencies below 0.1 ms demands beyond 5G's capabilities [20]. The 6G era, expected to emerge around 2030, seeks to address these requirements by not only increasing network performance but also enabling new paradigms of connectivity that integrate advanced technologies such as artificial intelligence (AI), machine learning (ML), and quantum communication into the core of the network [21].

3.2. Limitations of 5G in addressing future needs

Although 5G represents a major advancement over previous mobile generations, it still falls short of meeting the requirements of future intelligent communication systems. One fundamental limitation lies in

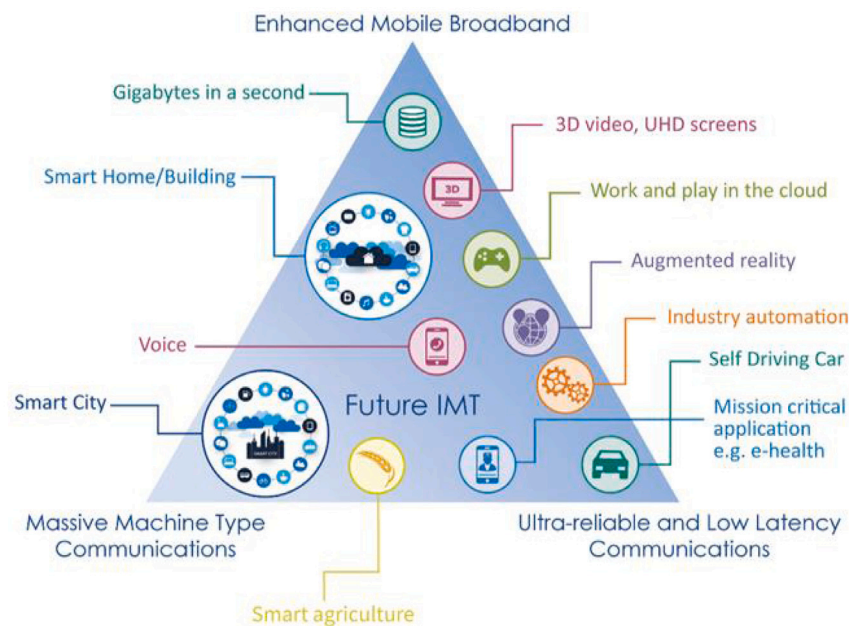


Fig. 3. The different 5G usage [17].

spectrum availability: even with the use of sub-6 GHz and millimeter-wave bands, 5G cannot provide the ultra-wide bandwidth necessary for emerging applications such as holographic telepresence, multi-user XR, and terabit-scale data services, which demand orders of magnitude more capacity than current deployments can offer [16,22]. Additionally, the dense small-cell infrastructure required for high-frequency operation significantly increases network energy consumption, raising sustainability concerns as device numbers continue to grow [23]. Another major constraint is 5G's inability to support ultra-massive connectivity at the scale expected for 6G, where billions of heterogeneous devices, including sensors, machines, autonomous robots, and intelligent surfaces, must coexist with strict reliability guarantees [24]. Latency is also a major bottleneck: despite achieving sub-millisecond performance under ideal conditions, 5G cannot consistently deliver the deterministic, end-to-end ultra-low latency required for real-time applications such as remote surgery, tactile Internet, or holographic communication [17]. Moreover, 5G lacks native support for AI-driven networking, limiting its ability to autonomously manage traffic flows, predict network behavior, or optimize resource allocation in complex, dynamic environments [25]. These limitations collectively illustrate the need for a new generation of wireless systems 6G, that can address spectrum scarcity, energy efficiency, connectivity density, intelligent operation, and extreme latency requirements.

3.3. Vision and objectives of 6G

The vision for 6G centers on creating an intelligent, ultra-reliable, and fully pervasive communication ecosystem capable of supporting the increasingly demanding applications of the next decade. Unlike 5G, which primarily enhances speed and connectivity, 6G aims to deliver terabit-per-second data rates, sub-0.1 ms latency, and globally integrated coverage by leveraging new spectrum ranges such as the sub-THz and THz bands [26]. A key ambition of 6G is to enable seamless end-to-end communication across terrestrial, aerial, satellite, and underwater systems, thereby ensuring ubiquitous connectivity even in remote or previously inaccessible regions [27]. Beyond raw performance improvements, 6G introduces a paradigm shift towards AI-driven network automation, where machine learning and distributed intelligence are embedded at every layer of the network to optimize resource management, predict traffic patterns, enhance reliability, and

enable autonomous operation. These capabilities are essential for supporting advanced and highly interactive applications such as extended reality (XR), autonomous and collaborative robotic systems, digital twins, and holographic communication, all of which require extreme data rates, deterministic latency, and real-time context awareness [28]. By integrating communication, sensing, computing, and AI into a unified framework, 6G aspires to create a transformative platform that not only extends human-to-human and human-to-machine interaction but also enables emerging intelligent ecosystems at global scale.

3.4. 6G timeline

Each generation of wireless communications technology typically spans around ten years before moving on to the next. 5G's journey began with its initial standard published in 2018 as part of 3GPP Release 15, marking the foundations of the current generation. Since then, 5G has evolved progressively, with subsequent releases introducing advanced capabilities. 3GPP Release 18, published in 2024, represent the next phase of 5G evolution, known as 5G Advanced. This phase aims to refine existing 5G functionalities while integrating new features to meet emerging demands [29].

Simultaneously, research and development efforts for the next-generation 6G systems are intensifying. The International Telecommunication Union (ITU), is expected to publish the IMT-2030 framework around 2026. This framework will outline the vision, objectives, and technical requirements for 6G, serving as a blueprint for future development.

Following the release of the IMT-2030 document, the 3rd Generation Partnership Project (3GPP), responsible for defining detailed specifications for mobile networks, will begin the development of the 6G standards. These specifications, designed to align with the ITU's vision, are projected to emerge between 2028 and 2030. This timeline is presented in Fig. 4 It aligns with the anticipated commercial deployment of 6G systems around 2030, ushering in a new era of wireless communication with unprecedented capabilities [30].

4. 6G enabling technologies

The realization of 6G communication networks will depend on a range of cutting-edge technologies designed to overcome the limitations of previous generations and meet the unprecedented demands



Fig. 4. Estimated timeline of standardization and delivery for 6G networks [31].

of the future. These enabling technologies are pivotal in achieving the extreme data rates, ultra-low latencies, and ultra massive connectivity envisioned for 6G systems. Below, we explore these technologies and their contributions.

4.1. Artificial intelligence and machine learning

AI is set to become one of the foundations of 6G networks, enabling intelligent decision-making and optimization across the network. From resource allocation to interference management, AI will empower 6G systems with self-learning and adaptive capabilities. Key applications include autonomous network configuration, anomaly detection, and predictive maintenance. AI facilitates enhanced user experience through personalized services and dynamic quality-of-service (QoS) adjustments. AI also enables closed-loop optimization to wireless 6G networks. In this technique, the user acts in accordance with device feedback. By maintaining end-to-end collective optimization at the system layer, AI can further be implemented to optimize wireless communication systems. This intelligent layer offers numerous possibilities, such as self-optimization, self-learning, and data collection. These examples are only a small part of the widespread applications of AI technologies and the innovations they will bring to the 6G network structure [32].

Machine learning, a subset of AI, will play an important role in analyzing and processing the massive amounts of data generated by 6G networks. ML algorithms will optimize traffic management, energy efficiency, and user mobility prediction. Reinforcement learning models will be used for real-time decision-making in scenarios such as spectrum allocation and interference mitigation [33].

4.2. Generative Artificial Intelligence (GenAI)

Generative Artificial Intelligence (GenAI) is poised to play a critical role in 6G, enabling the creation, prediction, and synthesis of complex network information using advanced generative models such as GANs, VAEs, diffusion models, and large language models (LLMs). Unlike traditional AI, which primarily performs classification or regression tasks, GenAI can generate realistic channel models, mobility patterns, network traffic predictions, synthetic training datasets, and even optimized communication waveforms [34].

In 6G physical-layer design, generative models enable the reconstruction of high-dimensional radio environments, allowing more accurate channel estimation and improved beamforming for THz and mmWave systems. GenAI also accelerates the creation of realistic digital twins by producing synthetic but reliable network states, thereby reducing data collection overhead [35].

At the network layer, GenAI facilitates proactive resource allocation, anomaly detection, and autonomous decision-making. Furthermore, generative models are essential for security in 6G, enabling adversarial training, synthetic attack simulation, and privacy-preserving data generation.

Recent publications demonstrate the potential of GenAI in communication systems, including intelligent waveform design, generative channel modeling, and zero-shot learning for network optimization. As 6G evolves towards AI-native architectures, GenAI will be fundamental for enabling self-evolving, self-healing, and self-optimizing networks [36].

4.3. Terahertz communication

The Terahertz (THz) band, presented in Fig. 5, spanning 0.1–10 THz, represents the last untapped region in the radio frequency spectrum, offering ultra-high bandwidth and data rates up to 1 Tbps. It bridges the gap between millimeter-wave (mmWave) and infrared (IR) frequencies, combining broader transmission bandwidth than mmWave with better propagation characteristics than IR. THz communication is essential for 6G, enabling applications such as holographic telepresence, and ultra-HD streaming [37].

With its ability to support ultra-dense networks and massive capacity, THz communication is poised to meet 6G's demands for unprecedented data rates and low latency. Advantages include higher spectral efficiency, integration of thousands of small antenna elements into base stations, and enhanced security through high directional beams. However, challenges like system complexity, power consumption, and THz component design must be addressed. Leveraging advancements like ultra-massive MIMO and Intelligent Reflective Surfaces, THz communication is a promising enabler for 6G networks [38].

4.4. Visual light communication

Visual Light Communication (VLC) uses visible light as a wireless communication medium, offering high data rates, immunity to electromagnetic interference, and secure, interference-free communication. It is especially suitable for indoor environments like hospitals and factories, and it complements traditional systems [39].

VLC operates similarly to fiber optics, delivering fiber-like transmission rates and bandwidth through light. With terahertz-level bandwidth, high-frequency reuse, and unlicensed spectrum, VLC supports gigabit-level transmission. However, current systems face challenges like narrow LED bandwidths and the need for efficient ultra massive MIMO designs.

Despite these hurdles, advancements in VLC are expected to enable terabit-level data rates for 6G by 2030, utilizing thousands of LED active sources for short-range, high-capacity communication [40].

The Fig. 6 shows an example of inside visible light communication.

4.5. Multi-access edge computing (MEC)

Multi-access Edge Computing (MEC) enhances network efficiency by bringing computational resources closer to end users, enabling low-latency, real-time processing for applications like autonomous vehicles and augmented reality (AR). By deploying computing, caching, and data analysis at the network edge, MEC significantly reduces end-to-end latency and optimizes resource management.

In 6G, MEC will integrate application-oriented capabilities into the network, supporting AI-driven services and creating dynamic, self-optimizing machine-type communication (MTC) networks (see next in Section 4.6). With features like intelligent data elimination and network slicing, MEC ensures flexibility, localized data processing, and energy-efficient operations. These advantages position MEC as an essential technology for achieving the performance and scalability demands of 6G networks [41].

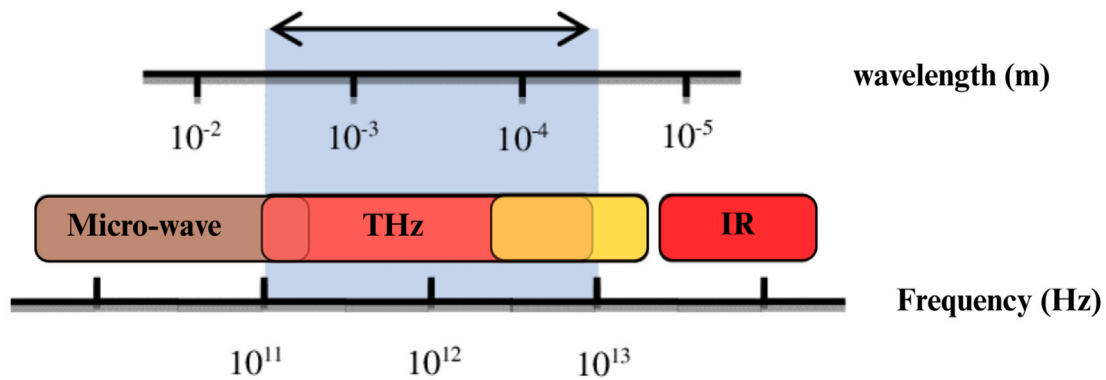


Fig. 5. The Terahertz (THz) band [38].

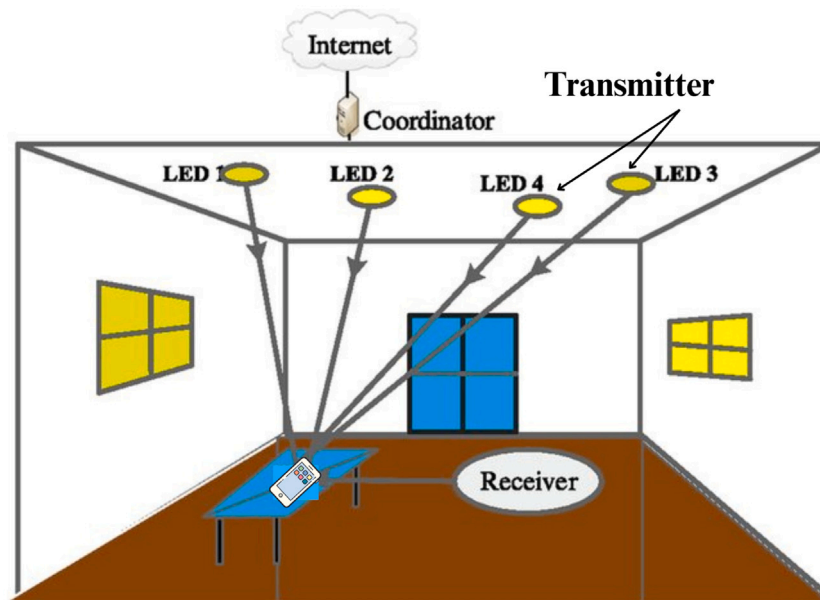


Fig. 6. Indoor visible light communication [40].

4.6. Machine type communication

6G is envisioned as the first wireless standard to offer seamless integration across diverse industry sectors, addressing the growing demand for reliable and scalable communication. It will facilitate digital transformation through vertical applications enabled by machine-type communication (MTC) [42]. MTC, which allows autonomous device-to-device communication, is crucial for IoT but poses challenges related to latency and reliability. Technologies such as Network Function Virtualization (NFV) and Software-Defined Networking (SDN) are used to address these challenges by providing flexible, efficient, and secure network infrastructures. A dynamic and unified core network will be essential for supporting the diverse requirements of 6G verticals. Furthermore, massive MTC (mMTC) will enable efficient low-power communication among billions of devices, reinforcing 6G's role as a foundational element for future intelligent networked systems [43].

4.7. Low earth orbit satellite communication

The integration of satellite capabilities with terrestrial networks in 6G is expected to address essential challenges such as efficient data transmission, resource optimization, disaster recovery, and global connectivity. Low Earth Orbit (LEO) satellites are key enablers of this vision, offering high-speed, low-latency communication and extended

coverage [44]. By supporting frequency reuse and enabling three-dimensional coverage in conjunction with terrestrial and aerial platforms, LEO networks enhance network scalability. Although satellite communication faces obstacles like delay and path loss, the adoption of mmWave technology provides high bandwidth and directivity for effective point-to-point links.

Emerging satellite architectures and heterogeneous networks further reduce deployment costs and improve coverage. Enabling technologies such as network slicing, NFV, and SDN support the seamless convergence of terrestrial and non-terrestrial infrastructures, enhancing automation, flexibility, and resource allocation. Edge computing reinforces this integration by improving mobility support, Quality of Service (QoS), and communication resilience. With all this advantage, LEO satellites play a foundational role in realizing the 6G vision of ubiquitous, reliable, and inclusive global connectivity [45].

4.8. Orbital angular momentum technique

Orbital Angular Momentum (OAM) is an emerging modulation technique that leverages the angular momentum of electromagnetic waves to transmit multiple data streams over the same frequency, thereby enhancing spectral efficiency and throughput. By utilizing helical phase wavefronts, OAM enables multiple access and robust frequency sharing,

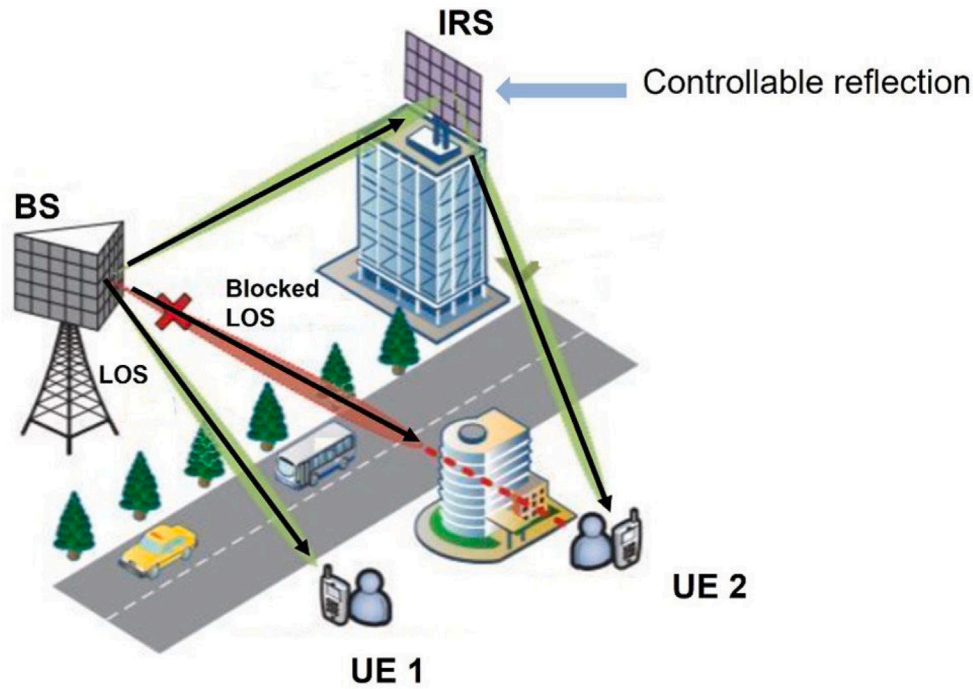


Fig. 7. Reconfigurable intelligent surfaces help ensure user equipment gets the highest received signal power [46].

making it a promising candidate for 6G networks [47]. Key advantages include high spectral efficiency, anti-jamming capabilities, and improved bandwidth utilization.

Despite its potential, OAM faces challenges such as beam divergence, which limits long-distance transmission, and sensitivity to misalignment, especially in non-line-of-sight (NLOS) conditions. Initially developed for optical systems, OAM has since been extended to Terahertz, mmWave, and lower frequency domains. Integration with advanced channel coding and frequency-sharing techniques is anticipated to enhance its viability. As research progresses, OAM is expected to be essential in overcoming bandwidth constraints and boosting 6G network performance [47].

4.9. Cognitive radio

Cognitive Radio (CR) is a dynamic spectrum access technology that enhances spectral efficiency by detecting and utilizing under-used frequency bands, thereby minimizing interference. In 6G, CR is expected to be instrumental in managing ultra-dense network deployments and maintaining seamless connectivity in complex and dynamic environments. Building on CR, Symbiotic Radio (SR) introduces a cooperative paradigm in which multiple subsystems within a heterogeneous network intelligently interact to optimize resource utilization and transmission efficiency [48].

While SR holds promise for improving overall network performance, it remains in its developmental stages, with challenges such as energy efficiency and effective spectrum optimization. The integration of CR, SR, and complementary technologies, such as inter-device communication, spectrum sharing, and full-duplex operation, will be important for enabling flexible and intelligent spectrum management in 6G systems.

4.10. Intelligent reflective surfaces

Intelligent Reflective Surfaces (IRS) are reconfigurable metasurfaces designed to enhance wireless communication by passively reflecting electromagnetic waves towards desired directions, as illustrated in Fig. 7. Composed of numerous low-cost reflective elements, IRS can

dynamically control the phase of incident signals without requiring active transmission or significant power consumption [49].

IRS significantly improves signal propagation in challenging scenarios such as non-line-of-sight (NLOS) and deep-fade environments by increasing the signal to interference and noise ratio (SINR). These surfaces can be strategically deployed on buildings, vehicles, and other structures to enhance coverage and reliability without modifying existing infrastructure. Applications include beam steering, beamforming, and multi-user communication, with performance further enhanced through integration with machine learning techniques such as deep learning and neural networks.

By complementing massive MIMO, a key 5G technology, IRS contributes to improved spectral efficiency and higher data rates, positioning it as an important component in future 6G networks [46].

4.11. Wireless energy transmission

Wireless Energy Transmission (WET) in 6G enables sustainable and wireless power delivery to devices like smartphones and IoT sensors. It reduces battery waste, extends device durability, simplifies maintenance, and lowers emissions. Integrated with Wireless Information and Energy Transmission (WIET), 6G base stations can transfer both data and electricity, supporting battery-free devices and enhancing IoT network efficiency and sustainability [50], as illustrated in Fig. 8.

4.12. Edge intelligence

Edge intelligence integrates AI with edge computing to enable real-time decision-making near data sources, reducing latency and bandwidth demands. This approach supports critical applications like autonomous vehicles, drones, augmented reality, and telesurgery, where milliseconds matter. Powered by 6G's ultra-low latency, high data rates, and scalability, edge intelligence enhances data privacy, computational efficiency, and seamless connectivity, marking a transformative advancement for intelligent systems. Fig. 9 presents the flowchart of edge computing [51].

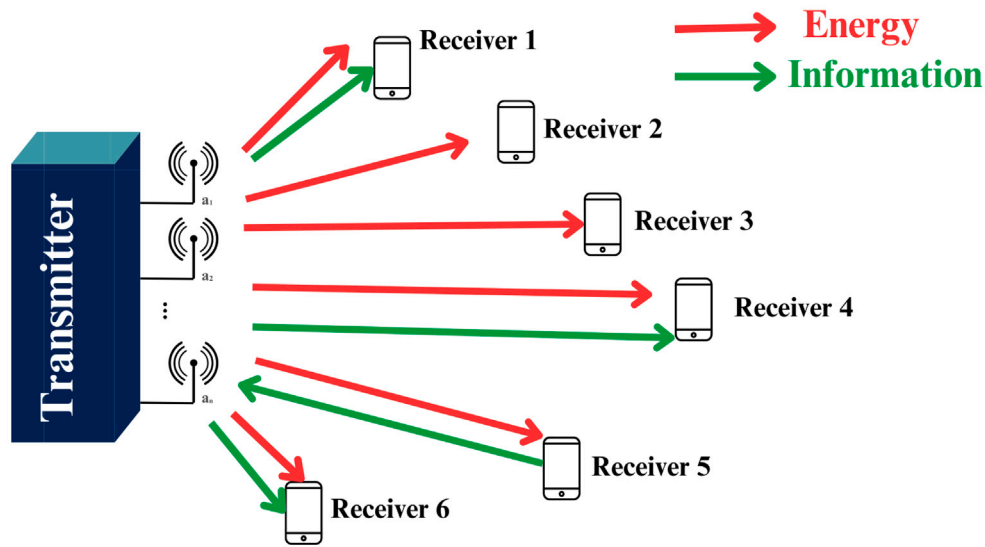


Fig. 8. Model for transfer of wireless information and energy.

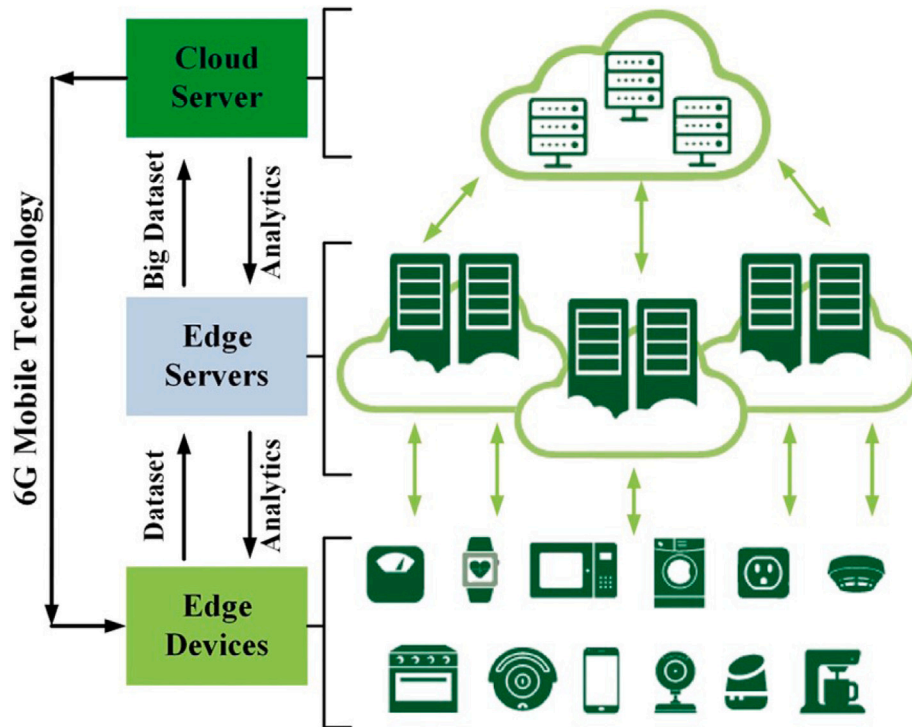


Fig. 9. Edge computing [51].

4.13. Digital twin networks

Digital Twin Networks (DTNs) are emerging as a transformative technology for 6G, enabling real-time mirroring, prediction, and optimization of physical network environments through virtual replicas. A digital twin integrates heterogeneous data sources, including sensors, network states, user mobility patterns, and environmental parameters, to maintain an accurate, continuously updated model of the network. This virtual environment can be used to test configurations, simulate failures, optimize resource allocation, and design proactive control strategies without affecting the live system. In 6G, DTNs will be essential for managing the increasing complexity introduced by

ultra-massive MIMO, 3D integrated networks (ground–air–space–sea), and AI-native communication layers. By offering predictive analytics and closed-loop control, DTNs enable self-adaptation, fault prediction, proactive load balancing, and end-to-end Quality of Service (QoS) optimization. Integrating DTNs with edge intelligence further ensures low-latency synchronization between the physical and digital domains [52]. Recent works highlight the relevance of DTNs in future networks. For instance, emerging frameworks propose DTN-enabled RAN optimization, virtual testing for UAV-assisted communication, and dynamic resource allocation for non-terrestrial networks. These studies confirm that DTNs will serve as a cornerstone in realizing fully autonomous, resilient, and intelligent 6G infrastructures [53].

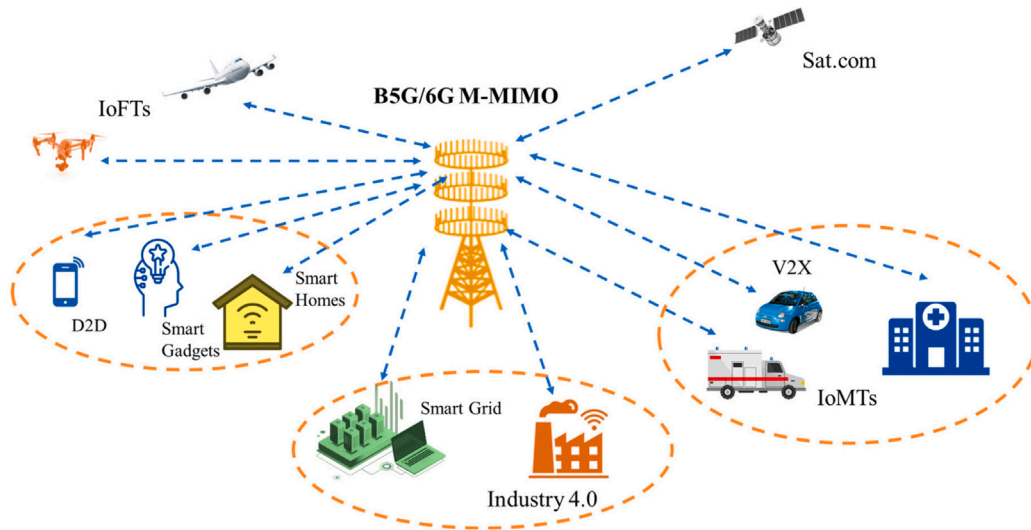


Fig. 10. Multi-antenna array-based massive MIMO for B5G/6G [55].

4.14. Free duplexing and spectrum sharing (FDSS)

Duplexing in 6G enables simultaneous data transmission and reception, overcoming inefficiencies of earlier techniques like FDD and TDD used in 1G–4G. With a full duplex system, 6G allows users to share all resources (frequency, time, and space) efficiently, achieving low latency and high throughput. Advanced spectrum sharing, supported by AI and blockchain, will enable dynamic resource management. Non-orthogonal multiple access (NOMA) further enhances spectrum efficiency, latency, and user capacity, making it a key technology for 6G communication systems [54].

4.15. Ultra massive MIMO

Ultra Massive MIMO, with its extensive antenna arrays, significantly enhances spectral efficiency and signal quality through spatial multiplexing and beamforming. In 6G, it will address capacity issues by enabling precise, high-directionality beams and mitigating interference, while also supporting high-frequency bands like THz. With over 10,000 antenna elements, 6G will achieve unprecedented spectrum efficiency and reduced propagation loss, ensuring high data rates and extended signal coverage [55]. Fig. 10 presents the Multi-Antenna Array-Based Massive MIMO for B5G/6G.

4.16. Blockchain technology

Blockchain technology provides decentralized, secure frameworks essential for managing 6G networks, addressing spectrum-sharing challenges, and enabling efficient resource utilization. Integrated with AI, it enhances QoS through smart resource allocation, scalable networks, and advanced caching. Blockchain's trust, transparency, and low-latency capabilities support secure data sharing, identity management, and transaction validation in multi-stakeholder ecosystems, making it a transformative tool for the telecommunications industry [56].

4.17. Tactile internet

The Tactile Internet enables real-time human-to-machine (H2M) interactions through haptic sensors, allowing users to remotely sense and control devices. With applications in fields like healthcare, virtual reality, and military operations, it relies on ultra-low latency and high reliability, which 6G technology is poised to deliver [57].

4.18. Quantum communication and quantum ML

Quantum computing (QC) enables parallel processing of large, multidimensional data, significantly reducing computation times for complex algorithms like successive interference cancellation (SIC). QC will enhance 6G capabilities by providing high computational power, advanced link capacity, and ultra-secure communication [58].

5. The new architecture for the core network

6G networks aim to provide energy-efficient and globally seamless wireless connectivity while addressing the limitations of current network architectures, which are inadequate for meeting the stringent requirements of future applications, such as ultra-high throughput, ultra-low latency, and enhanced reliability. Consequently, forward-looking research into next-generation network frameworks is essential. The FG NET-2030 initiative established Sub-Group 3 to design the architecture for Network 2030 [8]. However, accurately defining the future network architecture remains challenging. As a compromise, Sub-Group 3 has opted to interpret the architecture from multiple dimensions rather than proposing a unified framework. This section explores the architectural changes anticipated for 6G, as depicted in Fig. 11.

In addition to these early perspectives, recent advances highlight that 6G architecture will be fundamentally AI-native, modular, and fully integrated across terrestrial, aerial, space, and underwater domains. The architecture will rely on programmable, software-defined components and distributed intelligence to enable real-time optimization, predictive control, and autonomous operation. Emerging paradigms such as Digital Twin Networks (DTN) provide a virtual replica of the network that continuously synchronizes with physical elements, enabling proactive maintenance, fault prediction, and dynamic resource management. These digital replicas also support scenario testing and network optimization without disturbing live operations, significantly enhancing reliability and adaptability.

5.1. From terrestrial to ubiquitous 3D coverage

A key goal of next-generation network architecture is to extend communication coverage both horizontally and vertically. The current terrestrial cellular infrastructure has two major limitations: an inability to support high-altitude and deep-sea communication scenarios, which are critical for future services, and prohibitively high costs

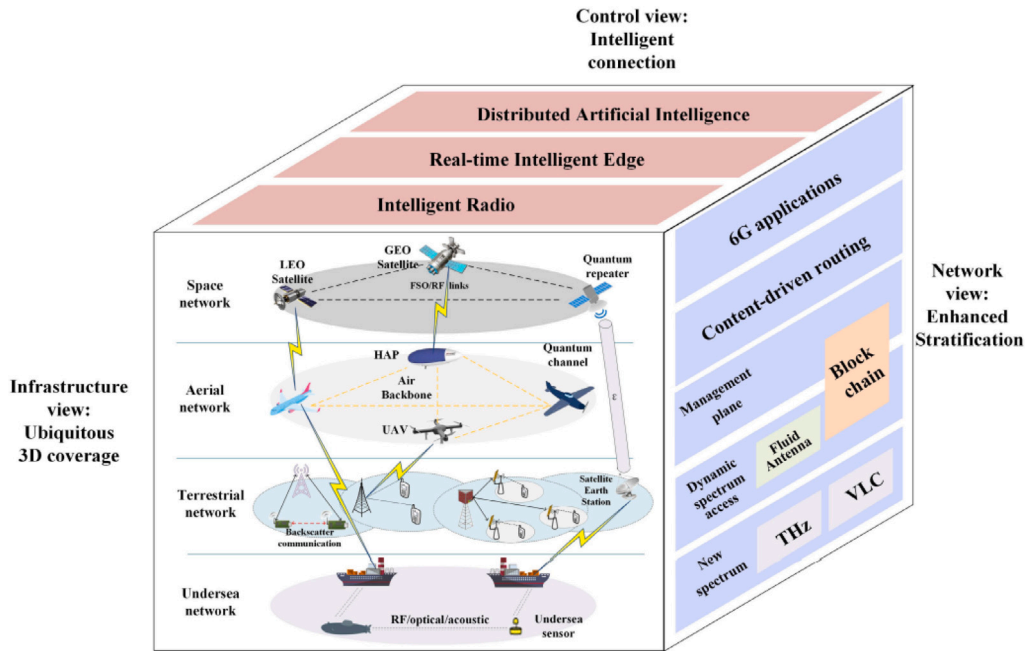


Fig. 11. Different dimensions of the architecture of 6G network [8].

associated with dense cellular networks required for global-scale connectivity. To address these limitations, 6G will integrate non-terrestrial networks to deliver comprehensive wireless coverage. So the initial concepts for Space–Air–Ground–Sea integrated communication have been discussed [59].

Beyond simple multi-layer integration, the 3D architecture of 6G must provide seamless handover across different platforms with varying mobility patterns, propagation characteristics, and link budgets. This requires new routing protocols, AI-assisted beamforming, cross-domain synchronization, and interoperability frameworks to maintain stable connectivity between satellites, HAPs, UAVs, terrestrial cells, and underwater systems.

5.1.1. Space network: LEO satellite systems

High-Throughput Satellite (HTS) systems now offer broadband Internet services with cost and bandwidth performance comparable to terrestrial networks. However, most communication satellites operate in geostationary orbit (GEO) at 35,786 km altitude, resulting in high latency and integration challenges with terrestrial mobile networks. To overcome these limitations, Non-Geostationary Orbit (NGSO) satellite systems — such as those in Low Earth Orbit (LEO) and Medium Earth Orbit (MEO) — have been proposed to provide global, low-latency, high-throughput Internet connectivity [60]. Major NGSO initiatives include: Starlink, with over 11,000 planned satellites by 2027 [8]; OneWeb, targeting a 720-satellite LEO constellation; and China's Hongyan project, expected to deploy 320 satellites by 2025.

Though large-scale deployment and integration of NGSO systems with terrestrial networks are still in progress, theoretical and simulation results demonstrate their potential. Notably, LEO satellite networks with laser and RF co-routing can outperform terrestrial fiber links in latency over distances greater than 3000 km.

To enable global connectivity, the Space-Terrestrial Integrated Network (STIN) architecture has been introduced, comprising space-based backbone networks (SBN), terrestrial networks (TN), and space-based access networks (SAN) [61]. In this model, GEO satellites (SBN) ensure wide-area coverage with inter-satellite links (ISLs), while LEO and MEO satellites (SAN) facilitate integration with HTS and terrestrial networks for ubiquitous access.

Emerging technologies play a key role in enhancing STIN functionality. Software-Defined Networking (SDN) and Information-Centric Networking (ICN) offer adaptive control, efficient configuration, and reduced latency. Additionally, Multipath TCP (MPTCP) ensures high throughput and robust mobility management through seamless handover capabilities, as validated in recent evaluations.

To ensure stable global connectivity, future LEO networks will be deeply integrated with terrestrial cores using SDN/NFV paradigms. AI-driven route prediction will mitigate Doppler effects and optimize inter-satellite link utilization, while Digital Twin models will help operators evaluate constellation performance, simulated failures, and congestion management strategies.

5.1.2. Aerial network: Flexible relay services

Aerial networks are generally divided into two primary categories: High-Altitude Platforms (HAPs), which operate in the stratosphere, and Low-Altitude Platforms (LAPs), which typically function at altitudes of a few kilometers [59].

- **High-Altitude Platforms (HAPs):** These systems provide broad coverage areas and long endurance, offering capabilities comparable to Low Earth Orbit (LEO) satellite networks.
- **Low-Altitude Platforms (LAPs):** This category includes Unmanned Aerial Vehicles (UAVs), which are characterized by rapid deployment, high maneuverability, and efficient short-range communication performance.

UAV-based LAPs can serve as relay nodes to support long-distance communication and facilitate seamless integration between terrestrial and non-terrestrial networks. Their flexibility and adaptability make them especially useful in emergency and disaster recovery situations where conventional communication infrastructure is either unavailable or severely damaged [60].

Furthermore, UAVs and HAPs may be equipped with reconfigurable intelligent surfaces (RIS) or lightweight THz transceivers, enabling aerial beam steering, temporary backhaul creation, and adaptive coverage extension. Their integration introduces new challenges such as flight-based mobility management, energy constraints, and airspace regulation, which must be addressed for safe and predictable operation.

Table 3

Comparison of underwater wireless communication technologies.

Parameter	RF	Acoustic	Optical
Attenuation	High	Lowest	Depends on turbidity
Achievable data rate	~Mbps	~Kbps	~Gbps
Latency	Moderate	High	Low
Transmission distance	<10 m	<100 km	<100 m
Power consumption	Moderate	High	Low

5.1.3. Undersea network

The potential inclusion of undersea networks in future 6G systems remains a topic of debate. Underwater wireless communication primarily utilizes three technologies: radio frequency (RF), acoustic, and optical communication [62]. A comparative analysis of these technologies is summarized in Table 3.

The deployment of undersea communication systems presents several significant challenges due to the highly unpredictable and dynamic nature of the underwater environment. Establishing stable network connectivity is inherently difficult, as water conditions can change rapidly with variations in temperature, salinity, turbidity, and ocean currents, all of which affect signal propagation and system performance. Moreover, underwater channels suffer from severe signal attenuation, particularly for radio frequency and optical communication, while acoustic signals, although capable of long-range transmission, experience low data rates and high latency. The extreme physical conditions of deep-sea environments, including high pressure, corrosion, and biofouling, further increase the risk of equipment degradation or failure. These factors make network setup, maintenance, and long-term operation considerably more complex than in terrestrial or aerial communication scenarios, highlighting the need for robust designs, adaptive communication techniques, and resilient hardware capable of withstanding harsh underwater conditions.

6. Use cases and emerging services

6G mobile technology introduces significant advancements, including high bandwidth, ultra-reliable communication, increased data rates, low latency, and an ideal environment for AI integration. These enhancements open up new application areas, ranging from military uses to meeting personal needs, particularly in scenarios requiring remote services, as highlighted during the pandemic. The demand for reliable communication with high data rates has become increasingly critical. Now we explore the opportunities and benefits of 6G technology, highlighting its practical applications. Some of these applications are presented in Fig. 12.

6.1. Connected Robotics and Autonomous Systems (CRAS)

The advancement of Collaborative Robotics and Autonomous Systems (CRAS) is a major driver in the development of 6G networks. CRAS includes emerging technologies such as autonomous vehicles, drone swarms, flying delivery systems, and robotic platforms, all of which impose stringent requirements on network performance beyond those addressed by traditional IoT services in 5G (example in Fig. 13) [63].

These applications demand ultra-low latency for real-time coordination and high throughput to handle the vast volumes of data generated by sensors, cameras, and control systems. For example, autonomous vehicles rely on seamless data exchange to enable synchronized navigation and ensure safety. Similarly, drone swarms used in defense, disaster response, or precision agriculture require rapid communication to operate collaboratively. Flying delivery systems also depend on stable, low-latency connectivity to ensure timely and efficient logistics operations [64]. By meeting these performance requirements, 6G will be a foundational enabler for next-generation CRAS deployments across a wide range of industrial and societal applications.

6.2. Extended reality

Extended Reality (XR), which encompasses Augmented Reality (AR), Mixed Reality (MR), and Virtual Reality (VR), represents a major advancement in immersive human-machine interaction. XR technologies merge physical and digital environments, enabling users to interact with computer-generated content in real time. Among them, Mixed Reality (MR) offers the most seamless integration, allowing dynamic interaction between real and virtual objects by eliminating the barriers that traditionally separate the two domains [65].

The realization of high-quality XR experiences requires ultra-low latency, high bandwidth, and reliable connectivity, capabilities that 6G is poised to deliver. Through its support for real-time, high-resolution content delivery and interactive holographic applications, 6G will unlock the full potential of XR technologies. Furthermore, the integration of edge computing and AI-driven network intelligence in 6G will ensure enhanced responsiveness, stability, and personalization of XR environments, making them suitable for applications across education, healthcare, entertainment, and industry.

6.3. Internet of Everything (IOE)

The Internet of Everything (IoE) extends the concept of the Internet of Things (IoT) by integrating people, data, processes, and devices into a unified intelligent network. Through interconnected sensing devices, IoE facilitates real-time identification, monitoring, and decision-making across diverse environments, generating actionable insights for smarter systems and services. These sensors are capable of capturing a broad array of physical and biological parameters, enabling applications in healthcare, smart cities, and industrial automation [66].

As IoE scales to billions of interconnected elements, it imposes stringent requirements on network performance, including scalability, reliability, and latency. Sixth-generation (6G) networks are positioned as a key enabler for IoE, offering ultra-reliable low-latency communication (URLLC) and enhanced mobile broadband (eMBB) to support real-time, high-throughput, and energy-efficient connectivity. Consequently, 6G will be essential in unlocking the full potential of IoE in future hyper-connected environments.

6.4. UAV based mobility

Unmanned Aerial Vehicles (UAVs), commonly known as drones, have been extensively utilized in defense and military operations for applications such as remotely operated aircraft and autonomous drones. Over the years, the versatility of UAVs has extended their use into civilian domains, including disaster relief, agricultural plantation protection, traffic monitoring, and environmental sensing. These applications have highlighted the potential of UAVs as indispensable tools in addressing critical societal and environmental challenges [67].

With the emergence of 6G and the Internet of Everything (IoE), UAVs are anticipated to evolve further into a UAV-to-Everything (U2X) network. This paradigm shift will enable UAVs to connect seamlessly with other devices and systems, expanding their utility beyond traditional communication. By leveraging advanced 6G capabilities such as ultra-low latency, high data rates, and massive connectivity, UAVs will optimize sensing applications and enhance their communication modes, unlocking their full potential for societal and industrial transformation [68].

6.5. Holographic telepresence

Holographic Telepresence (HT) is an emerging communication technology that enables the real-time projection of life-size, three-dimensional (3D) representations of people and objects, offering a sense of physical presence and immersion. By capturing, compressing, transmitting, and reconstructing visual information using high-speed



Fig. 12. Some applications envisaged for 6G [13].

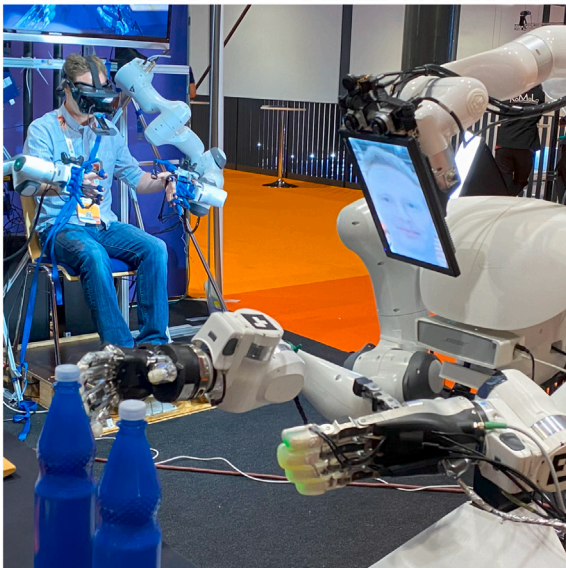


Fig. 13. Immersive teleoperation for connected robotic manipulation.

broadband and laser-based projection systems, HT facilitates applications such as 3D video conferencing, remote collaboration, and immersive content delivery [69].

The success of HT relies on ultra-high data rates and extremely low latency, which are achievable with the advent of 6G networks. With data transmission speeds of several Gbps and end-to-end latency as low as 100 μ s, 6G can support the stringent requirements of HT, including synchronized audio-visual rendering and real-time interactivity. Furthermore, the integration of tactile feedback and interactive technologies promises to elevate HT from visual immersion to full sensory engagement, marking a paradigm shift in remote communication and virtual experience design.

6.6. Wireless Brain-Computer Interactions (BCI)

The advent of 6G is poised to transform the healthcare sector through the support of advanced biomedical applications, including

smart body implants and wireless brain-computer interfaces (BCIs). These technologies surpass the capabilities of existing systems, which primarily focus on the control of basic prosthetics or isolated neural interfaces. By leveraging the ultra-low latency, massive device connectivity, and extreme reliability of 6G, future wireless BCIs will enable direct brain-to-device communication, facilitating precise control over prosthetics and seamless interaction with external digital systems. Concurrently, smart implants will provide continuous health monitoring and intelligent therapeutic interventions, enhancing patient care and enabling proactive treatment [70].

The integration of these systems imposes stringent performance demands on network infrastructure, comparable to, but more critical than, those of Extended Reality (XR) applications. In contrast to XR, where user immersion is central, BCIs directly affect human neural function and physiological health, thereby introducing the novel concept of Quality of Patient Experience (QoPE). Ensuring ultra-reliable communication, minimal delay, and fail-safe operation is thus essential to safeguard patient well-being and guarantee clinical efficacy.

6.7. Personalized body area networks

Body Area Networks (BANs), when integrated with mobile health (mHealth) systems, facilitate real-time, personalized health monitoring through continuous data acquisition from body-worn sensors. These networks enable dynamic communication with external systems, including social platforms, promoting proactive healthcare management. BANs support a wide range of medical applications such as chronic disease monitoring, early diagnosis, and simplified procedures like polysomnography by reducing reliance on cumbersome equipment [71]. Additionally, they extend to non-medical domains including emotion recognition, immersive entertainment, and biometric-based secure authentication, thereby broadening their impact across various sectors.

6.8. Intelligent healthcare

The emergence of Healthcare 5.0, propelled by advancements in artificial intelligence (AI) and digital wellness technologies, is set to revolutionize the healthcare sector. AI-driven smart healthcare systems will utilize Quality of Life (QoL) indicators and Intelligent Wearable Devices (IWD) to deliver personalized, efficient medical solutions [72]. Real-time health monitoring enabled by wearable sensors and edge

Internet of Healthcare (IoH) or future hospitals

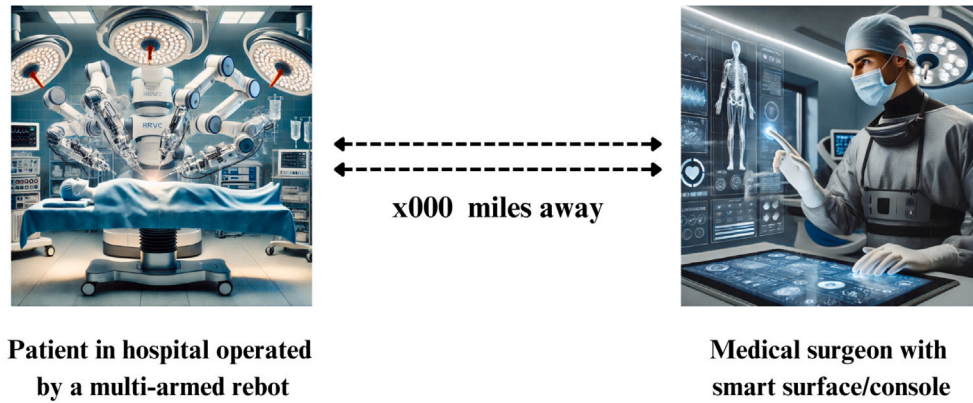


Fig. 14. Application for healthcare in 6G [73].

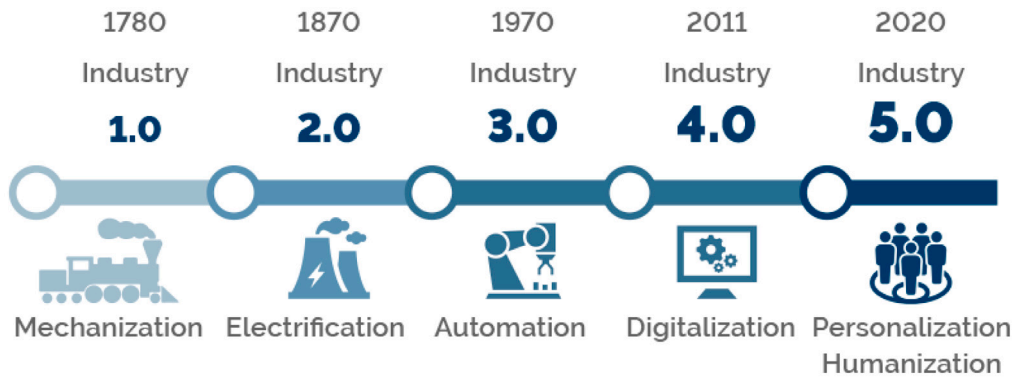


Fig. 15. Evolution of the industry.

computing allows for immediate data processing and remote diagnosis, thereby improving patient outcomes and enabling timely interventions [73]. The integration of 6G technologies, such as holographic communication, tactile internet, and intelligent robotics, further extends healthcare capabilities by supporting advanced applications like the remote surgery presented in Fig. 14. These innovations are especially very important in scenarios that demand reduced physical interaction, such as during global pandemics (like COVID-19).

6.9. Industry 5.0

Industry 5.0 marks the next phase of industrial evolution (Fig. 15), emphasizing synergistic collaboration between humans and intelligent machines. Building upon the core principles of Industry 4.0, automation, connectivity, and efficiency, it introduces a human-centric approach that integrates human creativity and intuition with robotic precision. This transformation is underpinned by 6G connectivity and the Internet of Everything (IoE), which will enable seamless interaction among a vast array of interconnected devices and systems. By leveraging these technologies, Industry 5.0 aims to deliver highly personalized and adaptive services while preserving operational efficiency, ultimately reshaping sectors such as manufacturing and logistics into more sustainable and intelligent ecosystems [74].

6.10. Use cases: Practical example, remote holographic surgery

To provide a clearer understanding of how 6G technologies will operate in real-world environments, we present a practical scenario demonstrating the combined use of holographic telepresence, tactile

Internet, AI-native processing, digital twins, and THz communication. This example illustrates the operational workflow, benefits, and challenges of implementing 6G-enabled systems.

In this scenario, a specialist surgeon located in a major hospital performs a real-time operation on a patient situated in a remote or underserved region using 6G-enabled surgical robotics. High-resolution volumetric imaging, multi-camera inputs, and biometric signals are transmitted through THz and optical wireless links to construct an accurate holographic representation of the patient in real time. The surgeon interacts with this hologram using haptic gloves supported by tactile-internet technologies, allowing precise manipulation of virtual tissues. At the remote site, robotic surgical instruments replicate the surgeon's movements with sub-millisecond latency to avoid motion distortions or delays. Edge computing guarantees instantaneous processing of sensory and control data, while a Digital Twin model continuously simulates each surgical action before execution, ensuring safety and optimizing instrument trajectories.

The implementation workflow of the system can be summarized as follows:

- **Sensing and Imaging:** 3D acquisition using high-resolution sensors, depth mapping, and medical imaging fusion.
- **Real-Time Data Transmission:** THz communication combined with RIS-assisted beamforming to maintain ultra-high-rate and ultra-low-latency links.
- **Haptic Control Loop:** Sub-millisecond bidirectional feedback enabling accurate and stable robotic manipulation.
- **AI-Assisted Decision Making:** Predictive models detect anomalies, estimate tissue deformation, and support surgical decision processes.

- **Digital Twin Simulation:** Each action is validated within a virtual replica of the surgical environment to prevent errors and optimize movement paths.
- **Execution:** The remote robotic system performs the surgical procedure while holographic and biometric feedback is continuously monitored by both the surgeon and AI algorithms.

This scenario highlights the transformative potential of 6G in healthcare. The main benefits include enabling access to specialized care in remote regions, improving surgical precision through AI-enhanced predictions and digital-twin validation, reducing human error, and supporting real-time collaboration between experts worldwide. However, several challenges remain, such as meeting extremely strict latency requirements (below 0.1 ms), ensuring the security and privacy of sensitive medical data, managing the computational complexity of hologram rendering, and guaranteeing network redundancy to prevent critical failures during surgical procedures.

7. Technical and theoretical challenges

The emergence of 6G mobile networks brings a plethora of opportunities, but it also introduces significant technical and theoretical challenges. These challenges must be addressed to meet the stringent requirements of ultra-high data rates, extremely low latency, massive connectivity, and seamless integration of terrestrial and non-terrestrial networks. Equalization techniques, which mitigate inter-symbol interference (ISI) and improving signal quality, offer promising solutions to some of these challenges [75].

7.1. Challenges in 6G mobile networks

The development and deployment of 6G mobile networks introduce a wide array of technical and engineering challenges due to their ambitious performance targets. Below are the major challenges expected in 6G networks

7.1.1. High-frequency communication and propagation loss

6G networks are projected to operate in the sub-terahertz (sub-THz) and terahertz (THz) frequency bands (100 GHz–10 THz) to support data rates exceeding 1 Tbps. However, electromagnetic waves in this frequency spectrum are highly susceptible to propagation losses due to atmospheric absorption (e.g., water vapor, oxygen molecules) and molecular scattering. These losses significantly limit the transmission range and coverage area. Additionally, such high-frequency signals experience increased vulnerability to multipath fading, non-line-of-sight (NLOS) conditions, and physical obstructions, necessitating the development of advanced beamforming, intelligent reflecting surfaces (IRS), and channel equalization techniques.

7.1.2. Ultra massive MIMO and channel complexity

To meet the demands of ultra-dense connectivity and spatial multiplexing, 6G will employ ultra-massive multiple-input multiple-output (UM-MIMO) systems, potentially involving thousands of antenna elements. While this allows for high spectral efficiency and energy-efficient communication, it also introduces considerable challenges in terms of channel estimation, beam alignment, pilot contamination, and hardware calibration. The increased dimensionality leads to high computational complexity and makes real-time signal processing and resource allocation significantly more difficult. Efficient and scalable algorithms, possibly leveraging machine learning, are necessary for practical implementation.

7.1.3. Low-latency and high-reliability requirements

Emerging applications such as holographic telepresence, real-time remote surgery, tactile internet, and autonomous vehicular systems impose ultra-low latency (below 1 ms) and extremely high reliability (up to 99.99999%). Maintaining these stringent quality-of-service (QoS) metrics is challenging under dynamic and error-prone wireless environments. Factors such as inter-symbol interference (ISI), synchronization errors, and fluctuating signal-to-noise ratio (SNR) must be effectively mitigated using robust modulation schemes, predictive resource scheduling, adaptive coding techniques, and robust equalization technique.

7.1.4. Integration of terrestrial and non-terrestrial networks

6G envisions a fully integrated communication framework that combines terrestrial networks with non-terrestrial components, including low Earth orbit (LEO) satellites, high-altitude platforms (HAPs), and unmanned aerial vehicles (UAVs). This heterogeneous architecture introduces new types of delays, Doppler shifts, and synchronization issues due to varied altitudes and velocities. Furthermore, differing channel characteristics (e.g., free-space vs. multipath-rich urban channels) require adaptive and cross-domain signal processing strategies for seamless interoperability and reliable service delivery.

7.1.5. Dynamic channel conditions

Due to high user mobility, frequent changes in network topology, and sensitivity to environmental conditions (e.g., rain, fog, or urban clutter), 6G wireless channels are expected to be highly non-stationary and dynamic. These variations result in time-varying ISI, Doppler effects, and unpredictable interference patterns. Consequently, traditional static channel models are inadequate, necessitating the adoption of real-time channel modeling, environment-aware communication protocols, and AI-based predictive algorithms to ensure robust performance.

8. Conclusion

The transition towards sixth-generation (6G) mobile networks represents an essential step in the ongoing evolution of wireless communication. Driven by the limitations of 5G and the exponential rise in demand for higher speeds, lower latency, and ubiquitous connectivity, 6G aims to redefine how we interact with digital technologies and the physical world. This paper has surveyed the fundamental enabling technologies, including artificial intelligence, terahertz communication, intelligent reflective surfaces, edge computing, and satellite integration, that are expected to power 6G systems. In doing so, it has highlighted how these innovations will address the growing needs of emerging applications such as extended reality, autonomous systems, holographic telepresence, and wireless brain-computer interfaces.

Beyond the technological vision, the paper has also addressed the significant theoretical and practical challenges that must be overcome, including spectrum limitations, energy efficiency, complex channel dynamics, and the need for intelligent, adaptive network architectures. These obstacles demand a multi-disciplinary approach, combining advancements in communication engineering, artificial intelligence, quantum computing, signal processing, and sustainable network design.

As we look towards the 2030 horizon, 6G is poised to become more than just a faster network, it will be the foundation of a truly intelligent, interconnected, and immersive digital ecosystem. Continued research and collaboration across academia, industry, and government will be essential in shaping the future of 6G and unlocking its transformative potential for society.

Looking ahead, several research directions are expected to play a decisive role in shaping the evolution of 6G networks. A primary avenue involves the development of AI-native architectures and protocols capable of embedding intelligence, self-adaptation, and predictive decision-making directly into the core of the communication system.

Addressing the limitations of high-frequency propagation will also require continued exploration of hybrid THz–optical communication models to support reliable, scalable terabit-class data transmission. Moreover, realizing seamless global coverage will depend on advanced cross-layer optimization strategies for space–air–ground–sea integrated networks, enabling intelligent spectrum management, efficient resource allocation, and robust mobility support across heterogeneous platforms. Energy efficiency remains a critical priority, motivating further innovation in ultra-massive MIMO, RIS-assisted communication, and low-power hardware design to ensure sustainable large-scale deployments. Security and privacy concerns will likewise demand attention, particularly through the development of quantum-resistant cryptographic schemes, blockchain-enabled trust mechanisms, and privacy-preserving learning frameworks. Finally, real-world prototypes and testbeds, especially for holographic communication, remote surgery, and brain–computer interfaces, will be essential to validate theoretical concepts and accelerate the practical realization of 6G technologies. Collectively, these directions underscore the interdisciplinary nature of 6G research and highlight the continued need for innovation across communication engineering, artificial intelligence, computing, and emerging application domains.

CRedit authorship contribution statement

Sofiane Latreche: Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Hocine Bellahsene:** Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] J.B. Liu, X. Zhang, J. Cao, L. Chen, Mean first-passage time and robustness of complex cellular mobile communication network, *IEEE Trans. Netw. Sci. Eng.* 11 (3) (2024) 3066–3076.
- [2] M. Shafi, R.K. Jha, S. Jain, 6G: Technology evolution in future wireless networks, *IEEE Access* 12 (2024) 57548–57573.
- [3] T.H. SVu, S.K. Jagatheesaperumal, M.D. Nguyen, N. Van Huynh, S. Kim, Q.V. Pham, Applications of generative AI (GAI) for mobile and wireless networking: A survey, *IEEE Internet Things J.* (2024).
- [4] Y. Goh, S. Oh, Y. Kim, J.M. Chung, Handover delay analysis of standalone and non-standalone 5G mobile networks, *IEEE Wirel. Commun.* (2025).
- [5] S. Jahandar, I. Shayea, E. Gures, A.A. El-Saleh, M. Ergen, M. Alnakhl, Handover decision with multi-access edge computing in 6G networks: A survey, *Results Eng.* (2025) 103934.
- [6] S.I. Loutfi, I. Shayea, U. Tureli, A.A. El-Saleh, W. Tashan, An overview of mobility awareness with mobile edge computing over 6G network: Challenges and future research directions, *Results Eng.* (2024) 102601.
- [7] F. Dong, F. Liu, Y. Cui, S. Lu, Y. Li, Sensing as a service in 6G perceptive mobile networks: Architecture, advances, and the road ahead, *IEEE Netw.* (2024).
- [8] T. Huang, W. Yang, J. Wu, J. Ma, X. Zhang, D. Zhang, A survey on green 6G network: Architecture and technologies, *IEEE Access* 7 (2019) 175758–175768.
- [9] M.A. Uusitalo, P. Rugeland, M.R. Boldi, E.C. Strinati, P. Demestichas, M. Ericson, et al., 6G vision, value, use cases and technologies from European 6G flagship project Hexa-X, *IEEE Access* 9 (2021) 160004–160020.
- [10] R. Bassoli, F.H. Fitzek, E.C. Strinati, Why do we need 6G? *ITU J. Futur. Evol. Technol.* 2 (6) (2021) 1–31.
- [11] A.I. Salameh, M. El Tarhuni, From 5G to 6G—challenges, technologies, and applications, *Futur. Internet* 14 (4) (2022) 117.
- [12] S.A.A. Hakeem, H.H. Hussein, H. Kim, Vision and research directions of 6G technologies and applications, *J. King Saud Univ. Comput. Inf. Sci.* 34 (6) (2022) 2419–2442.
- [13] S. Latreche, H. Bellahsene, A. Taleb-Ahmed, Some applications envisaged for the new generation of communications networks 6G, in: *The 1st National Conference on Electronics, Electrical Engineering, Telecommunications, and Computer Vision*, 2023, pp. 1–6.
- [14] KUMAR, Amit, LIU, Yunfei, SENGUPTA, Jyotsna, et al., Evolution of mobile wireless communication networks: 1G to 4G, *Int. J. Electron. Commun. Technol.* 1 (1) (2010) 68–72.
- [15] HANDARI, Nikhil, DEVRA, Shivinder, et SINGH, Karamdeep, Evolution of cellular network: from 1G to 5G, *Int. J. Eng. Tech.* 3 (5) (2017) 98–105.
- [16] D. Hui, S. Sandberg, Y. Blankenship, M. Andersson, L. Grosjean, Channel coding in 5G new radio: A tutorial overview and performance comparison with 4G LTE, *IEEE Veh. Technol. Mag.* 13 (4) (2018) 60–69.
- [17] M.H.C. Garcia, A. Molina-Galan, M. Boban, J. Gozalvez, B. Coll-Perales, T. Şahin, A. Kousaridas, A tutorial on 5G NR V2X communications, *IEEE Commun. Surv. Tutor.* 23 (3) (2021) 1972–2026.
- [18] P. Pirinen, A brief overview of 5g research activities, in: *1st International Conference on 5G for Ubiquitous Connectivity*, IEEE, 2014, pp. 17–22.
- [19] P. Varga, Á.I. Jászberényi, D. Pásztor, B. Nagy, M. Nasar, D. Raisz, How beyond-5G and 6G makes IoT and the smart grid green—a survey, *Sensors* 25 (13) (2025) 4222.
- [20] V. Raj, C.A. Ancy, Understanding the future communication: 5G to 6G, *Int. Res. J. Adv. Sci. Hub* 3 (6S) (2021) 17–23.
- [21] M.S. Akbar, Z. Hussain, M. Ikram, Q.Z. Sheng, S.C. Mukhopadhyay, On challenges of sixth-generation (6G) wireless networks: A comprehensive survey of requirements, applications, and security issues, *J. Netw. Comput. Appl.* 233 (2025) 104040.
- [22] A.A. Ateya, A.A. Abd El-Latif, A. Muthanna, A. Volkov, A. Koucheryavy, Enabling Metaverse and Telepresence Services in 6G Networks, CRC Press, 2025.
- [23] N. Hu, Z. Tian, X. Du, M. Guizani, An energy-efficient in-network computing paradigm for 6G, *IEEE Trans. Green Commun. Netw.* 5 (4) (2021) 1722–1733.
- [24] P. Nidagundi, A. Darvekar, M. Amber, A comparative study of ultra-massive MIMO intelligent receivers with adversarial robustness and energy efficiency for 6G applications, *Results Eng.* (2025) 105409.
- [25] N. Koursiompas, S. Barmounakis, I. Stavrakakis, N. Alonistioti, AI-driven, context-aware profiling for 5G and beyond networks, *IEEE Trans. Netw. Serv. Manag.* 19 (2) (2021) 1036–1048.
- [26] A. Al-jzari, Y. He, J. Hu, S. Salous, Sub-THz and THz channel measurements and characteristic analysis in indoor and outdoor environments for 6G wireless systems, *IEEE Trans. Terahertz Sci. Technol.* (2025).
- [27] N. Saeed, H. Almorad, H. Dahrouj, T.Y. Al-Naffouri, J.S. Shamma, M.S. Alouini, Point-to-point communication in integrated satellite-aerial 6G networks: State-of-the-art and future challenges, *IEEE Open J. Commun. Soc.* 2 (2021) 1505–1525.
- [28] H.F. Ahmad, W. Rafique, R.U. Rasool, A. Alhumam, Z. Anwar, J. Qadir, Leveraging 6G, extended reality, and IoT big data analytics for healthcare: A review, *Comput. Sci. Rev.* 48 (2023) 100558.
- [29] P. Hung, H. Kan, G. Knopf, Overview of 5G and 6G, 2025.
- [30] R. Liu, R.Y.N. Li, M. Di Renzo, L. Hanzo, A vision and an evolutionary framework for 6G: Scenarios, capabilities and enablers, 2023, arXiv preprint arXiv:2305.13887.
- [31] SP-231044 : 3GPP TSG#101 Bengaluru, India, sept 11 15, 2023.
- [32] M.R. Mahmood, M.A. Matin, P. Sarigiannidis, S.K. Goudos, A comprehensive review on artificial intelligence/machine learning algorithms for empowering the future IoT toward 6G era, *IEEE Access* 10 (2022) 87535–87562.
- [33] M.A. Matin, S.K. Goudos, S. Wan, P. Sarigiannidis, E.M. Tentzeris, Artificial intelligence (AI) and machine learning (ML) for beyond 5G/6G communications, *EURASIP J. Wirel. Commun. Netw.* 2023 (1) (2023) 22.
- [34] Y. Gao, Z. Lu, Y. Wu, Y. Jin, S. Zhang, X. Chu, et al., Enabling 6G through multi-domain channel extrapolation: Opportunities and challenges of generative artificial intelligence, *IEEE Commun. Mag.* (2025).
- [35] R. Zhong, X. Mu, M. Jaber, Y. Liu, Enabling distributed generative artificial intelligence in 6G: Mobile edge generation, *IEEE Internet Things J.* (2024).
- [36] Y.J. Liu, H. Du, X. Xu, R. Zhang, G. Peng, B. Cao, et al., A survey of integrating generative artificial intelligence and 6G mobile services: Architectures, solutions, technologies and outlooks, *IEEE Trans. Cogn. Commun. Netw.* (2025).
- [37] S. Latreche, H. Bellahsene, A. Taleb-Ahmed, Terahertz for radar applications and wireless communication, in: *The First National Conference on Advances in Computational Intelligence, Systems and Networking*, 2023, pp. 1–6.
- [38] S. Latreche, H. Bellahsene, A. Taleb-Ahmed, Terahertz for wireless communication and radio frequency, in: *The 2nd Electrical Engineering International Conference, EEIC'23*, 2023, pp. 1–6.
- [39] E. Niarchou, A.C. Boucouvalas, Z. Ghassemloooy, L.N. Alves, S. Zvanovec, Visible light communications for 6g wireless networks, in: *2021 Third South American Colloquium on Visible Light Communications, SACVLC, IEEE*, 2021, pp. 01–06.
- [40] S. Naser, L. Bariah, S. Muhaidat, P.C. Sofotasios, M. Al-Qutayri, E. Damiani, M. Debbah, Toward federated-learning-enabled visible light communication in 6G systems, *IEEE Wirel. Commun.* 29 (1) (2022) 48–56.
- [41] L. Zhao, G. Zhou, G. Zheng, I. Chih-Lin, X. You, L. Hanzo, Open-source multi-access edge computing for 6G: Opportunities and challenges, *IEEE Access* 9 (2021) 158426–158439.
- [42] N.H. Mahmood, S. Böcker, I. Moerman, O.A. López, A. Munari, K. Mikhaylov, et al., Machine type communications: key drivers and enablers towards the 6G era, *EURASIP J. Wirel. Commun. Netw.* 2021 (1) (2021) 134.

- [43] Y. Li, J. Dai, Z. Si, K. Niu, C. Dong, J. Lin, et al., Unsourced multiple access for 6G massive machine type communications, *China Commun.* 19 (3) (2022) 70–87.
- [44] X. Lin, S. Cioni, G. Charbit, N. Chuberre, S. Hellsten, J.F. Boutillon, On the path to 6G: Embracing the next wave of low earth orbit satellite access, *IEEE Commun. Mag.* 59 (12) (2021) 36–42.
- [45] S. Aliaga, V. Petrov, J.M. Jornet, Modeling interference from millimeter wave and Terahertz bands cross-links in low earth orbit satellite networks for 6G and beyond, *IEEE J. Sel. Areas Commun.* (2024).
- [46] M.A.S. Sejan, M.H. Rahman, B.S. Shin, J.H. Oh, Y.H. You, H.K. Song, Machine learning for intelligent-reflecting-surface-based wireless communication towards 6G: A review, *Sensors* 22 (14) (2022) 5405.
- [47] H. Zhang, Z. Cao, H. Xie, H. Jin, Orbital angular momentum (OAM) in wireless communication: applications and challenges towards 6G, in: 2023 14th International Conference on Information and Communication Technology Convergence, ICTC, IEEE, 2023, pp. 842–847.
- [48] M.M. Aslam, L. Du, X. Zhang, Y. Chen, Z. Ahmed, B. Qureshi, Sixth generation (6G) cognitive radio network (CRN) application, requirements, security issues, and key challenges, *Wirel. Commun. Mob. Comput.* 2021 (1) (2021) 1331428.
- [49] Q. Liu, S. Sun, B. Rong, M. Kadoch, Intelligent reflective surface based 6G communications for sustainable energy infrastructure, *IEEE Wirel. Commun.* 28 (6) (2021) 49–55.
- [50] Y. Che, Z. Zhao, S. Luo, K. Wu, L. Duan, V.C. Leung, UAV-aided wireless energy transfer for sustaining internet of everything in 6G, *Drones* 7 (10) (2023) 628.
- [51] B. Yang, X. Cao, K. Xiong, C. Yuen, Y.L. Guan, S. Leng, et al., Edge intelligence for autonomous driving in 6G wireless system: Design challenges and solutions, *IEEE Wirel. Commun.* 28 (2) (2021) 40–47.
- [52] M. Sheraz, T.C. Chuah, Y.L. Lee, M.M. Alam, A.A. Al-Habashna, Z. Han, A comprehensive survey on revolutionizing connectivity through artificial intelligence-enabled digital twin network in 6G, *IEEE Access* 12 (2024) 49184–49215.
- [53] X. Lin, L. Kundu, C. Dick, E. Obiodu, T. Mostak, M. Flaxman, 6G digital twin networks: From theory to practice, *IEEE Commun. Mag.* 61 (11) (2023) 72–78.
- [54] V. Ziegler, H. Viswanathan, N. Batra, How 6G will enable the industrial metaverse, 2024, Authorea Preprints.
- [55] S. Latreche, H. Bellahsene, A. Taleb-Ahmed, Optimizing 6g networks: Impact of massive MIMO on throughput in 6G communications, in: S. Bououden, S. Ziani, I. Chadli (Eds.), *Proceedings of the 6th International Conference on Electrical Engineering and Control Applications – Volume 1, ICEECA 2024*, in: *Lecture Notes in Electrical Engineering*, vol. 1471, Springer, Singapore, 2026, http://dx.doi.org/10.1007/978-981-95-1109-9_43.
- [56] H.H. Pajooh, S. Demidenko, S. Aslam, M. Harris, Blockchain and 6G-enabled IoT, *Inventions* 7 (4) (2022) 109.
- [57] G.P. Fettweis, H. Boche, 6G: The personal tactile internet—and open questions for information theory, *IEEE BITS Inf. Theory Mag.* 1 (1) (2021) 71–82.
- [58] M.Z. Ali, A. Abohmra, M. Usman, A. Zahid, H. Heidari, M.A. Imran, Q.H. Abbasi, Quantum for 6G communication: A perspective, *IET Quantum Commun.* 4 (3) (2023) 112–124.
- [59] M.A. Jamshed, A. Kaushik, S. Manzoor, M.Z. Shakir, J. Seong, M. Toka, et al., A tutorial on non-terrestrial networks: Towards global and ubiquitous 6G connectivity, 2024, arXiv preprint [arXiv:2412.16611](https://arxiv.org/abs/2412.16611).
- [60] N. Heydarishahreza, T. Han, N. Ansari, Spectrum sharing and interference management for 6G LEO satellite-terrestrial network integration, *IEEE Commun. Surv. Tutor.* (2024).
- [61] M. Sheng, D. Zhou, W. Bai, J. Liu, H. Li, Y. Shi, J. Li, Coverage enhancement for 6G satellite-terrestrial integrated networks: performance metrics, constellation configuration and resource allocation, *Sci. China Inf. Sci.* 66 (3) (2023) 130303.
- [62] Y. Ma, Q. Zhang, H. Wang, 6G: Ubiquitously extending to the vast underwater world of the oceans, *Engineering* 8 (2022) 12–17.
- [63] L. Qiao, Y. Li, D. Chen, S. Serikawa, M. Guizani, Z. Lv, A survey on 5G/6G, AI, and robotics, *Comput. Electr. Eng.* 95 (2021) 107372.
- [64] X. Wang, Y. Guo, Y. Gao, Unmanned autonomous intelligent system in 6G non-terrestrial network, *Information* 15 (1) (2024) 38.
- [65] H. Yu, M. Shokrnezhad, T. Taleb, R. Li, J. Song, Toward 6g-based metaverse: Supporting highly-dynamic deterministic multi-user extended reality services, *IEEE Netw.* 37 (4) (2023) 30–38.
- [66] P.K. Padhi, F. Charrua-Santos, 6G enabled industrial internet of everything: Towards a theoretical framework, *Appl. Syst. Innov.* 4 (1) (2021) 11.
- [67] S. Latreche, H. Bellahsene, A. Taleb-Ahmed, Optimizing UAV trajectory for 5G/6G service support during disaster events using k-means clustering and rotation-based algorithm, in: 2024 International Conference on Advances in Electrical and Communication Technologies, ICAECOT, IEEE, 2024, pp. 1–6.
- [68] K. Messaoudi, O.S. Oubbati, A. Rachedi, A. Lakas, T. Bendouma, N. Chaib, A survey of UAV-based data collection: Challenges, solutions and future perspectives, *J. Netw. Comput. Appl.* 216 (2023) 103670.
- [69] D. Svechnikov, R. Dunaytsev, A. Muthanna, A. Aziz, Holographic images delivery model toward 6g telepresence services, in: *International Conference on Next Generation Wired/Wireless Networking*, Springer Nature Switzerland, Cham, 2023, pp. 338–345.
- [70] H. Hu, Z. Wang, X. Zhao, R. Li, A. Li, Y. Si, et al., A survey on brain-computer interface-inspired communications: Opportunities and challenges, *IEEE Commun. Surv. Tutor.* (2024).
- [71] C. Sandeepa, B. Siniarski, N. Kourtellis, S. Wang, M. Liyanage, A survey on privacy of personal and non-personal data in B5G/6G networks, *ACM Comput. Surv.* 56 (10) (2024) 1–37.
- [72] S. Nayak, R. Patgiri, 6G communication technology: A vision on intelligent healthcare, *Health Inform.: A Comput. Perspect. Heal.* (2021) 1–18.
- [73] A. Ahad, Z. Jiangbina, M. Tahir, I. Shayea, M.A. Sheikh, F. Rasheed, 6G and intelligent healthcare: Taxonomy, technologies, open issues and future research directions, *Internet Things* (2024) 101068.
- [74] P.K.R. Maddikunta, Q.V. Pham, B. Prabadevi, N. Deepa, K. Dev, T.R. Gadekallu, et al., Industry 5.0: A survey on enabling technologies and potential applications, *J. Ind. Inf. Integr.* 26 (2022) 100257.
- [75] C. Li, Z. Liu, Z. He, Experimental demonstration of weak signal detection using photo-counting receiver with inter-symbol-interference, in: 2023 Asia Communications and Photonics Conference/2023 International Photonics and Optoelectronics Meetings, ACP/POEM, IEEE, 2023, pp. 1–3.